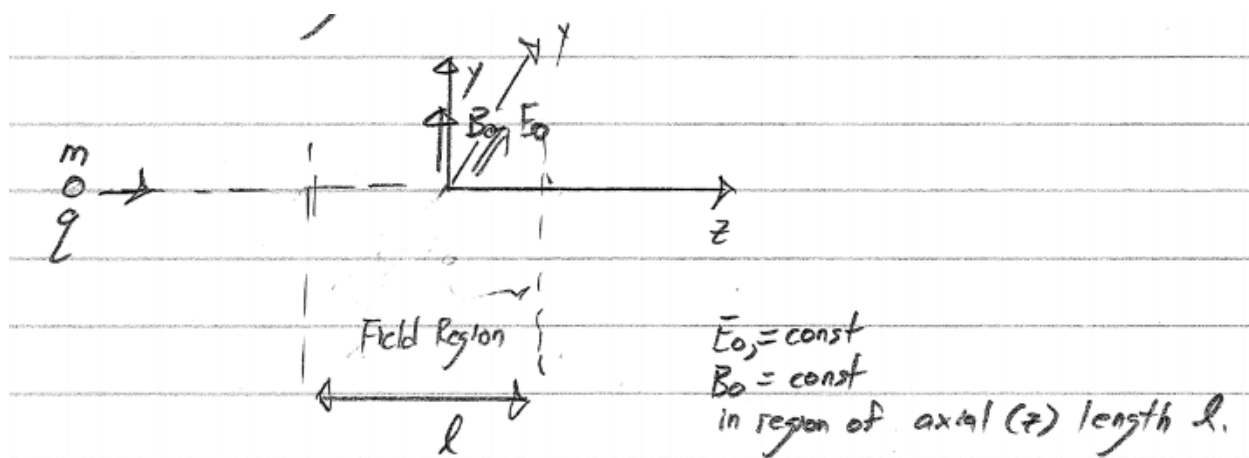


Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Charge to Mass Selector 30 pts.

Positively charged ions with charge q and mass m are emitted from a source and accelerated through a potential difference V_0 . The particles then enter a “velocity selector” made up of crossed electric and magnetic fields, E_0 and B_0 . Model these fields as uniform over an axial length, ℓ .



- 5 pts: Find a formula for the value of axial β such that the particle will move through the selector without deflection.
- 3 pts: What value of β will pass through the selector undeflected for $E_0 = 75$ kV/m and $B_0 = 0.05$ T?
- 5 pts: Assume non-relativistic motion. Show that if the magnetic field is turned off ($B_0 = 0$) that particles are deflected through an angle θ emerging from the deflector, with θ satisfying:

$$\tan(\theta) = \frac{1}{2} \frac{E_0 \ell}{V_0}$$

- 2 pts: If $\ell = 0.15$ m, $V_0 = 60$ kV, and $E_0 = 75$ kV/m evaluate θ in part (c).
- 5 pts: Assume the electric field E_0 and magnetic field B_0 are in balance for no deflection of the particles. Then the electric field is set to zero ($E_0 = 0$) and the magnetic field is left at its original value B_0 . Show that the particles will be bent on a circular arc of radius:

$$R = \frac{2V_0}{E_0}$$

- 2 pts: Find the value of R for $V_0 = 60$ kV and $E_0 = 75$ kV/m.

- g) 3 pts: Show for undeflected particles that the charge-to-mass ratio is:

$$\frac{q}{m} = \frac{E_0^2}{2V_0 B_0^2}$$

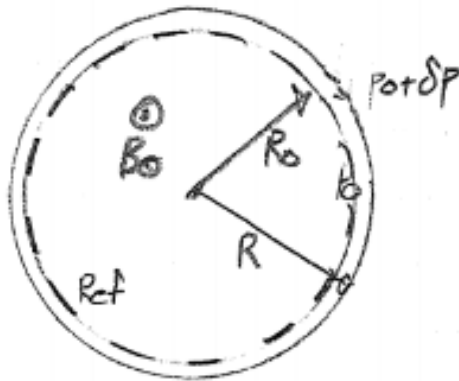
If ions have charge state $Q = q/e$ and atomic mass $A = m/m_u$, where $m_u = 931 \text{ MeV}/c^2$ is the rest mass of a nucleon, this value of q/m corresponds to:

$$\frac{Q}{A} = \frac{m_u c^2}{2eV_0} \left(\frac{E_0}{cB_0} \right)^2 = \frac{m_u c^2}{2eV_0} \beta^2$$

- h) 5 pts: Use the results and parameters above to find the value of Q/A for undeflected ions. If we believe those ions to be nitrogen, what charge state is most likely?

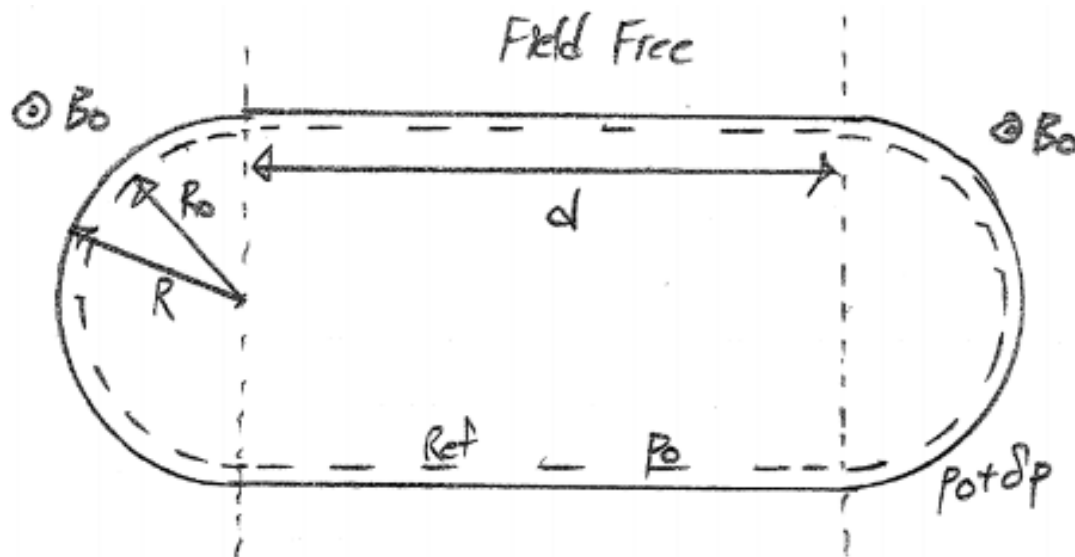
Problem 2. Slip Factor 20 pts.

- a) 5 pts: Consider a circular accelerator/storage ring composed of a uniform magnetic field, $\mathbf{B}_y = B_0 \hat{y} = \text{constant}$. The ideal reference path for a particle of momentum p_0 has radius R_0 in the plane perpendicular to $\mathbf{B}_y$. A particle with momentum $p = p_0 + \delta p$ will have a different closed path and radius R .



Calculate the slip factor, η , in terms of γ for this situation.

- b) 5 pts: Next, repeat part (a) for a racetrack accelerator with two uniform dipole bends separated by a field free drift of length d . Calculate the slip factor, η , in terms of γ and d/R_0 .



- c) 10 pts: For $d = 2R_0$ in part (b), plot η as a function of γ and note where it changes sign. Is this the “transition γ ”? What speed in $\beta = v/c$ does this correspond to?

Problem 3. Magnetic field from two straight wires 20 pts.

Two infinitely long, straight wires each carry a current I in the $-x$ direction. The wires are separated by a distance $2a$ in the y direction.

- 15 pts: Find an expression for the total magnetic field from the two wires at a point on the z axis halfway between the two wires, $(0, 0, z)$.
- 10 pts: Find the $z > 0$ value where the field is maximized.
- 5 pts: Draw a sketch of the field lines in the y - z plane for the magnetic field produced by the two wires.